



Life-Cycle Material Selection for Large-Scale Solar Concentrators: Performance, Durability, Energy Yield, and Techno-Economic Sustainability

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Abstract: Large-scale concentrated solar power (CSP) and concentrating photovoltaic (CPV) systems require material choices that simultaneously satisfy optical and thermal performance, structural reliability, environmental durability, and long-term economic viability. Material selection therefore affects not only initial plant cost but also degradation, maintenance demand, replacement frequency, lifetime energy yield, and levelized cost of energy (LCOE). This review examines the life-cycle performance of reflective, structural, receiver, absorber, and protective materials used in large-scale solar concentrators and evaluates how material properties interact with operating environment, energy performance, and economic sustainability. A structured narrative review with systematic elements was conducted using literature from major scientific databases and authoritative energy sources. Evidence on optical and thermal performance, degradation, service life, manufacturability, environmental exposure, life-cycle cost, and carbon implications was critically synthesized. A reduced-order energy-yield model and a normalized life-cycle sensitivity framework were also applied to clarify how degradation, operation and maintenance (O&M), replacement, discount rate, and project life influence material-related LCOE. The reviewed evidence showed that silvered-glass mirrors generally retained the strongest long-term durability record, with initial reflectivity of at least 93% and typical service lives of 20–30 years, whereas aluminium reflectors and polymer films offered lower mass and potential cost advantages but greater sensitivity to corrosion, ultraviolet exposure, abrasion, and replacement frequency. Low-cost absorber materials were found to be more suitable below approximately 300–400 °C, while high-temperature CSP applications required selective coatings and thermally stable receiver materials. The sensitivity analysis showed that material degradation, replacement burden, financing conditions, and project lifetime could substantially alter life-cycle cost comparisons, indicating that low purchase price alone is an inadequate basis for material selection. The review further showed that dust, humidity, salinity, ultraviolet radiation, thermal cycling, and cleaning practices strongly influence material suitability across different deployment environments. These findings indicate that material selection for large-scale solar concentrators should be based on retained energy performance, durability, climate exposure, maintenance requirements, and life-cycle cost rather than on initial material cost alone. A system-level strategy combining high-retention optical materials, lightweight structures, temperature-appropriate receivers, multifunctional protective coatings, climate-specific field validation, and transparent techno-economic assessment provides a more robust basis for reducing LCOE and supporting the long-term sustainability of CSP and CPV deployment.

Keywords: Concentrated solar power; Concentrating photovoltaics; Solar concentrators; Reflective materials; Receiver materials; Material degradation; Levelized cost of energy; Life-cycle assessment

1 Introduction

The global power sector faces the dual challenge of meeting growing electricity demand while accelerating decarbonization. In 2024, renewable sources supplied about one-third of global electricity and, together with nuclear power, accounted for roughly two-fifths of total electricity generation, while annual renewable-capacity additions

reached a record level. The International Renewable Energy Agency (IRENA) reported that global renewable power capacity reached 4,448 GW by the end of 2024, following the addition of 585 GW during that year. These developments reflect the continuing shift toward low-carbon electricity, although fossil fuels still account for a substantial share of global power generation [1–3].

Solar energy occupies a central position in this transition because of its broad geographical availability and its ability to meet both electricity and thermal-energy demand. Techno-economic studies of utility-scale solar conversion have shown that project feasibility depends on the combined effects of solar-resource availability, conversion efficiency, financing conditions, and operating assumptions rather than on nominal system performance alone [4]. Broader assessments of the green-energy transition have similarly shown that the economic case for renewable deployment becomes more complete when environmental impacts and the avoided consequences of fossil-fuel use are included in the evaluation [5].

Within the wider solar-energy portfolio, concentrator technologies used in concentrated solar power (CSP) and concentrating photovoltaic (CPV) systems increase incident solar irradiance over a smaller receiver or photovoltaic area. This concentration enables high-temperature heat generation, thermal-energy storage, and the use of high-efficiency photovoltaic cells [6, 7]. Concentrating technologies are also being investigated as alternatives to fossil-derived thermal energy and as potential “solar fuel” pathways for electricity generation and industrial applications [8, 9].

Material selection is a major determinant of both the technical and economic performance of solar concentrator systems because reflective surfaces, support structures, receivers, tracking components, and protective coatings account for a considerable part of system cost and long-term performance. These components must retain their optical, thermal, mechanical, and chemical properties over extended operating periods while being exposed to ultraviolet radiation, thermal cycling, wind loading, humidity, salinity, and airborne dust. Concentrator optics and structural systems can represent a substantial fraction of installed cost, which means that material selection directly affects project viability and the levelized cost of energy (LCOE) [10]. In CPV systems, the economic justification for using relatively expensive high-efficiency photovoltaic cells likewise depends on the availability of low-cost optical components and reliable thermal-management systems [11].

Conventional silvered-glass reflectors and galvanized-steel structures are technically mature and supported by extensive operating experience, but their high mass can increase transport requirements, foundation demand, structural loads, and embodied energy. These limitations have encouraged the development and use of aluminium reflectors, polymer-based reflective films, fibre-reinforced composites, bio-based structural materials, and lower-cost absorber surfaces [12, 13]. However, reductions in initial material cost do not necessarily translate into lower lifetime cost. Materials that degrade rapidly, require intensive cleaning, or undergo repeated replacement during a 20- to 30-year project lifetime may ultimately increase maintenance expenditure, reduce energy yield, and weaken the economic case for deployment [14].

Recent regional and application-oriented studies further show that concentrator performance cannot be evaluated independently of local solar-resource conditions and end-use requirements. Representative applications include parabolic-dish hybrid systems, CSP modelling using the System Advisor Model (SAM), integrated solar-thermal electricity and water-production systems, and linear Fresnel configurations for desalination [8, 9]. Linear Fresnel systems have also been examined specifically for solar-assisted desalination, illustrating the importance of matching collector geometry, material characteristics, and thermal-duty requirements within an integrated system design [4, 5].

Against this background, the present review evaluates material selection for large-scale solar concentrators from a life-cycle perspective. The analysis integrates optical performance, structural requirements, degradation behaviour, climatic exposure, energy-yield effects, maintenance demand, carbon implications, manufacturability, social considerations, and economic decision variables. Rather than seeking a universally low-cost material, the review aims to determine the conditions under which a material becomes technically and economically preferable once durability, operating environment, lifetime energy output, maintenance requirements, and life-cycle cost are considered together. This perspective provides a direct link between material performance and the long-term energy, economic, and environmental sustainability of large-scale CSP and CPV systems.

2 Methodology

This review adopted a structured narrative approach with systematic elements because the literature on solar-concentrator materials spans several closely connected fields, including materials science, optics, thermal engineering, structural mechanics, environmental durability, and energy economics. The available evidence comprises experimental studies, field observations, techno-economic analyses, simulations, review articles, and application-specific case studies. A quantitative meta-analysis was not considered appropriate because the reported performance metrics, exposure periods, test protocols, system boundaries, and economic assumptions were too heterogeneous to support direct statistical pooling.

2.1 Strategy of Literature Search

Relevant literature was identified through Scopus, Web of Science, ScienceDirect, IEEE Xplore, publishers' databases, and reports from authoritative energy agencies. The search strategy combined terms related to concentrator technologies and component functions with terms describing durability, economics, and environmental performance. Search terms included "solar concentrator materials", "CSP mirrors", "parabolic trough materials", "heliostat reflectors", "CPV optics", "receiver coatings", "structural materials", "degradation", "soiling", "corrosion", "life-cycle cost", "LCOE", "carbon emissions", and "environmental externalities". Backward reference tracking was also applied to key review papers and recent case studies in order to identify additional relevant sources.

2.2 Inclusion and Exclusion Criteria

Studies were included when they reported material properties or field performance relevant to solar concentrators; quantified degradation, environmental exposure, or service life; described scalable manufacturing or structural implications; or examined energy, economic, life-cycle, or environmental consequences. Studies addressing broader renewable-energy systems were retained only when they provided transferable methods or concepts relevant to sensitivity analysis, carbon valuation, social assessment, or transition planning. Studies dealing exclusively with non-CPV systems were excluded from the core material comparison unless they provided material-performance or environmental insights that could reasonably be transferred to concentrator applications.

2.3 Screening and Selection

Study selection was carried out in two stages. Titles and abstracts were first screened for relevance, followed by full-text assessment of the retained publications. Greater weight was given to studies containing field data, longer exposure periods, clearly reported optical or mechanical performance metrics, and explicit links between material behaviour and system-level performance. Recent regional studies were also retained when they provided relevant evidence on arid-climate resource uncertainty, solar-system modelling, or deployment constraints.

2.4 Data Extraction and Synthesis

For each retained study, information was extracted on material type, functional role, optical, thermal, and mechanical properties, environmental exposure conditions, dominant degradation mechanisms, service-life evidence, fabrication or installation requirements, and cost-related indicators. The extracted evidence was then organized by component function, including reflective materials, structural materials, receiver and absorber materials, and protective coatings. Cross-study findings were synthesized comparatively, with particular attention to the relationships among material performance, degradation, maintenance requirements, energy yield, and life-cycle cost. Economic comparisons were interpreted qualitatively or on a normalized basis because reported prices and cost assumptions varied across regions, publication years, suppliers, deployment scales, and financing conditions.

2.5 Climatic Data, Resource Modelling, and Validation Context

This review did not generate a new site-specific meteorological dataset and therefore did not attempt to validate a single climatic time series for any particular project location. Instead, it examined how climatic variables affect material performance and considered published approaches to solar-resource validation and modelling uncertainty. Solar-resource atlases can support regional screening, while transposition-model studies have shown that the choice of irradiance model can introduce uncertainty that subsequently propagates into energy-yield and economic estimates [15]. Cross-city studies in the Middle East and North Africa (MENA) region have similarly shown that the most suitable diffuse-transposition formulation can vary by location [5].

For arid and semi-arid applications, both climatic suitability and model-form uncertainty should therefore be considered when comparing concentrator materials. Recent analyses have shown that differences among irradiance-transposition models can materially affect predicted solar resources and downstream performance estimates [4]. Case studies from Libya have further demonstrated that project viability depends on local resource conditions, dust exposure, temperature, and technology assumptions rather than on nominal irradiance alone [8]. Reviews of solar-thermal deployment under Indian climatic conditions have likewise emphasized the need for climate-specific technology qualification [16].

2.6 Assumptions, Limitations, and Uncertainty

The synthesis assumed that the material properties reported in the reviewed studies were representative of the stated experimental or operating conditions. Cross-study comparisons were therefore interpreted as indicative ranges rather than as exact rankings. Sources of uncertainty include heterogeneous test protocols, extrapolation from accelerated-ageing experiments, differences in the definitions and measurement of reflectivity and absorptance, incomplete reporting of cleaning practices, regional cost variation, and uncertainty in future prices. Service-life

estimates for emerging polymers, coatings, and composite materials remain particularly uncertain because long-term field datasets are still limited.

The energy and sensitivity calculations presented later in the review are intentionally dimensionless and illustrative rather than predictive of a specific plant. Their purpose is to show how material degradation, maintenance requirements, replacement assumptions, financing conditions, and project lifetime can propagate into energy-yield and life-cycle cost metrics. This distinction avoids false precision while retaining the usefulness of the framework for comparing material-related decision variables.

3 Overview of Solar Concentrator Technologies

Solar concentrators increase incident solar flux by directing radiation onto a smaller receiver or photovoltaic area. Their concentration ratio, optical configuration, tracking requirements, and operating temperature determine the demands placed on reflector materials, structural accuracy, receiver performance, and thermal management [17, 18].

Line-focus systems, including parabolic troughs and linear Fresnel reflectors, typically operate at intermediate temperatures and rely on single-axis tracking. Point-focus systems, such as parabolic dishes and central-receiver towers, can operate at temperatures above 800 °C and consequently impose more stringent requirements on optical accuracy, structural stiffness, and receiver stability [18, 19]. Parabolic-dish systems have also been integrated with bioenergy in hybrid configurations, demonstrating that concentrating technologies can serve applications beyond stand-alone solar electricity generation [8].

Applications of solar concentrators extend across industrial process heat, electricity generation, desalination, and enhanced oil recovery. Parabolic-trough steam systems have been investigated for thermal enhanced-oil-recovery applications, while solar-thermal configurations have been developed to integrate electricity generation with water desalination [20]. These applications differ considerably in operating temperature, exposure to corrosive or fouling environments, thermal duty, and operating cycles. Material suitability and cost-effectiveness are therefore application dependent, since a material that performs adequately under moderate-temperature conditions may not provide the required optical, mechanical, or thermal stability under higher concentration ratios or more severe operating environments.

Table 1 compares the principal solar-concentrator configurations and the material challenges associated with each technology. The comparison shows that increasing concentration and operating temperature generally impose tighter requirements on optical accuracy, structural rigidity, tracking precision, and receiver thermal stability. These differences provide the technological basis for the material comparisons developed in the subsequent sections.

Table 1. Comparative characteristics of major solar concentrator technologies and associated material challenges

Type of Concentrator	Typical Configuration	Concentration Ratio	Operating Temperature	Tracking Requirement	Key Material Challenges
Parabolic-trough	Line-focus reflector	30–80×	300–450°C	Single-axis	Large mirror area, structural weight, reflector durability
Linear Fresnel reflector	Line-focus segmented mirrors	10–40×	250–400°C	Single-axis	Reduced optical efficiency, mirror alignment, low-cost reflectors
Solar tower (Heliostat field)	Point-focus central receiver	300–1000×	500–1000°C	Dual-axis	High-precision mirrors, high-temperature receivers, wind loads
Parabolic-dish	Point-focus dish concentrator	500–2000×	>800°C	Dual-axis	Structural stiffness, receiver thermal stability
Fresnel lens/compound parabolic concentrator	Static-or quasi-static optics	2–50×	<250°C	None or limited	Polymer degradation, ultraviolet stability, thermal ageing

The geometric differences summarized in Table 1 are illustrated schematically in Figure 1 through representative line-focus, point-focus, and static concentrator configurations. The comparison highlights how reflector area, support geometry, concentration arrangement, and receiver location redistribute material requirements among the optical, structural, and receiver subsystems.

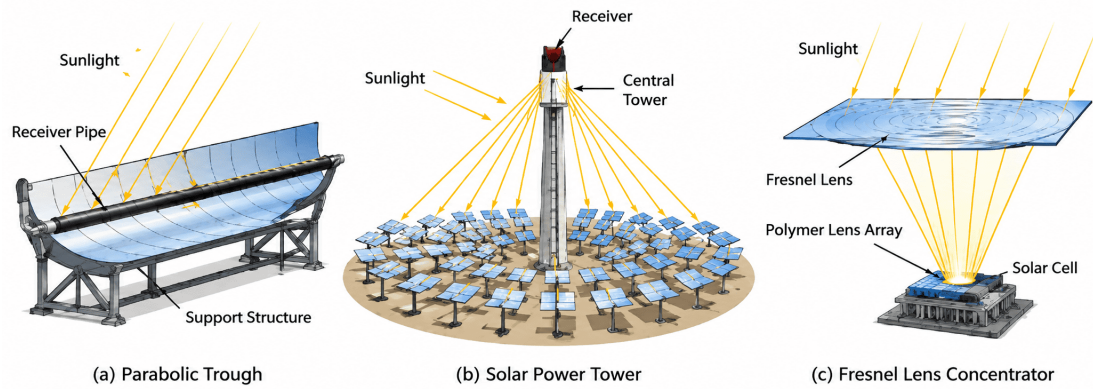


Figure 1. Schematic representation of major solar concentrator geometries: (a) line-focus parabolic trough; (b) point-focus solar tower with heliostat field; and (c) static Fresnel-based concentrator

4 Material Performance and Life-Cycle Assessment

4.1 Reflective Materials for Solar Concentrators

Reflective materials form the optical core of CSP concentrators and directly influence the amount of incident solar energy delivered to the receiver. Suitable reflector materials must provide high solar-weighted reflectance while maintaining geometric stability and surface integrity under prolonged exposure to ultraviolet radiation, thermal cycling, wind, humidity, salts, and abrasive dust. Even relatively small losses in reflectivity can become significant at field scale because they reduce the solar energy captured over the operating life of the system [19].

Silvered glass remains the commercial benchmark for large-scale solar concentrators. Low-iron glass incorporating a silver reflective layer and protective backing can provide initial solar-weighted reflectance above 93% and service lives of approximately 20–30 years when adequate edge protection and appropriate cleaning practices are maintained [11]. This established durability is accompanied by several disadvantages, particularly high mass, fragility, greater foundation requirements, transport burden, and handling costs. These factors can increase the structural and installation requirements of large concentrator fields despite the strong optical performance of the material.

Aluminium reflectors offer lower mass and can reduce structural loading and tracking inertia, although their initial reflectance is generally lower than that of silvered glass and their long-term performance depends strongly on oxide control and the integrity of protective surface layers [21]. In humid or saline environments, deterioration of these protective layers can accelerate corrosion and lead to progressive optical losses. Their economic attractiveness therefore depends not only on initial material and structural savings but also on the retention of reflective performance under the intended operating conditions.

Polymer reflective films provide substantial weight reduction and are compatible with scalable manufacturing processes such as roll-to-roll production. Their principal limitations are ultraviolet ageing, moisture ingress, abrasion, thermal distortion, and comparatively shorter field life. Barrier layers and ultraviolet (UV)-stabilized formulations can improve performance retention, although long-term field evidence remains less established than that available for silvered glass [22, 23]. Uncertainty is greater in dust-prone environments, where repeated particle deposition, cleaning, and abrasion can accelerate surface deterioration [24].

Table 2 compares the principal reflector options and illustrates a recurring life-cycle trade-off. Lightweight and potentially lower-cost alternatives can reduce structural and installation burdens but generally carry greater uncertainty regarding long-term optical retention and replacement requirements. Silvered glass, by contrast, imposes higher mass-related structural and installation costs but is supported by the strongest long-term durability evidence. Reflector selection should therefore account for retained optical performance, environmental exposure, maintenance requirements, and expected service life rather than initial material cost alone.

Figure 2 summarizes the dominant degradation pathways associated with the reflector materials considered in this review. Silvered-glass mirrors are particularly susceptible to edge corrosion and deterioration of protective backing layers, while aluminium reflectors depend on effective surface protection against oxidation and corrosion. Polymer-based reflective films are more sensitive to ultraviolet ageing, moisture ingress, and abrasive damage. These distinct degradation pathways help explain why reflector performance and service life depend strongly on both material architecture and operating environment.

Table 2. Comparison of commonly used and emerging reflective materials for solar concentrator systems

Reflective Material	Typical Reflectivity	Density/Weight	Expected Service Life	Key Advantages	Main Limitations
Silvered glass mirror	$\geq 93\%$	High	20–30 years	Excellent optical performance, proven durability	High weight, fragility, higher installation cost
Aluminium reflector	85–90%	Medium	10–20 years	Lightweight, lower structural cost	Oxidation, lower reflectivity
Polymer reflective film	88–92% (initial)	Very low	5–10 years	Ultra-lightweight, low manufacturing cost	Ultraviolet degradation, abrasion, limited lifetime

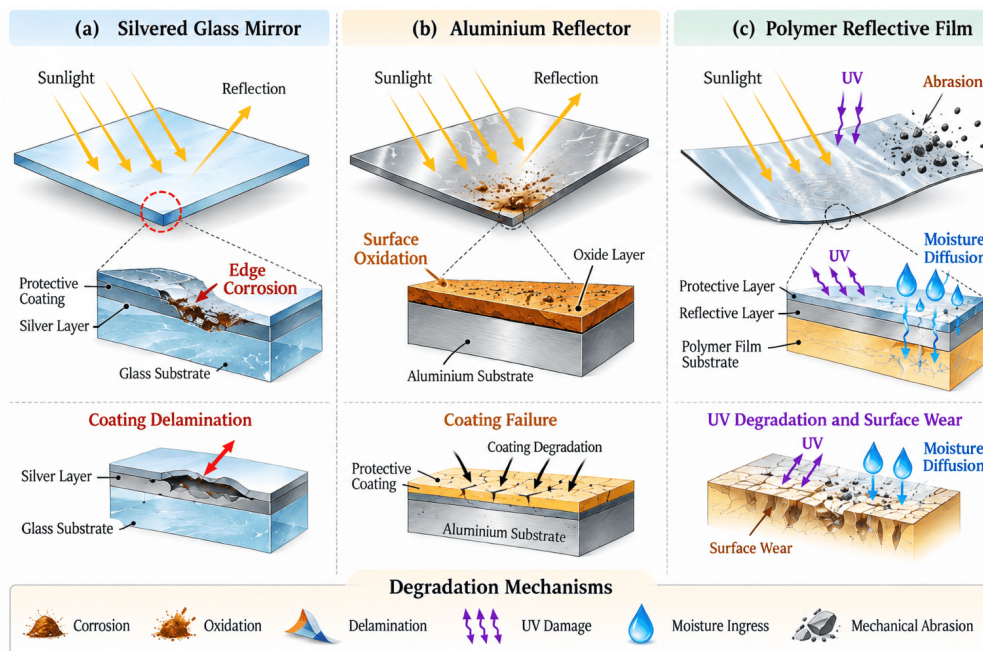


Figure 2. Schematic illustration of the dominant degradation mechanisms of reflective materials used in solar concentrator systems: (a) edge corrosion and coating delamination in silvered-glass mirrors; (b) surface oxidation and coating failure in aluminium reflectors; and (c) ultraviolet degradation, moisture ingress, and abrasive surface wear in polymer-based reflective films

Note: UV = ultraviolet.

4.2 Structural Materials for Solar Concentrator Support Systems

Support structures maintain the optical alignment of solar concentrators under self-weight, wind loading, thermal expansion, tracking motion, and repeated operating cycles. Their economic implications cannot be assessed from material price per unit mass alone. Material density influences the total structural mass, foundation requirements, transport and handling, and the loads imposed on tracking systems, while elastic modulus and strength determine the section dimensions required to maintain adequate stiffness and optical accuracy. Structural components can therefore account for a substantial share of the installed cost of concentrator systems [25, 26].

Galvanized carbon steel remains the conventional choice for many large-scale support structures because it combines high stiffness and strength with mature fabrication methods, widespread availability, and established engineering practice. Its principal disadvantages arise from its relatively high density and the need for corrosion protection, particularly under humid or coastal conditions [21]. The low material cost per unit strength must therefore

be considered alongside the additional structural mass, foundation requirements, and corrosion-management needs associated with long-term operation.

Aluminium alloys provide substantial weight reduction and easier handling while offering good resistance to corrosion. Their lower elastic modulus, however, generally requires larger structural sections to achieve stiffness comparable to steel, while the raw-material cost can also be higher [27]. The value of aluminium therefore depends on whether reductions in structural mass, transport requirements, foundation demand, and tracking loads compensate for the higher material cost and increased section dimensions.

Fibre-reinforced polymers (FRPs) and bio-based composites offer low density, favourable strength-to-weight ratios, and resistance to corrosion, making them potentially attractive for modular structures and applications in which reduced foundation loading or easier transportation is important. Their wider adoption is constrained by uncertainties related to creep, fatigue, fire behaviour, joining methods, standardization, and the limited availability of multi-decade field-performance data [28, 29]. These uncertainties are particularly important for large-scale concentrators, where small changes in structural geometry or stiffness can affect optical alignment over extended operating periods.

Table 3 compares the principal structural-material options considered in this review. Steel retains clear advantages in stiffness, engineering familiarity, and predictable structural performance, whereas aluminium, FRPs, and bio-based materials become increasingly relevant when reduced mass, corrosion resistance, transportation, or local manufacturing are important design considerations. The comparison reinforces the need to evaluate structural materials at the system level rather than on raw-material cost alone.

Table 3. Comparison of structural materials for large-scale solar concentrator support systems

Structural Material	Density	Mechanical Performance	Corrosion Resistance	Cost Considerations	Key Challenges
Galvanized steel	High	Very high stiffness and strength	Moderate (with coatings)	Low cost per unit strength	High weight, corrosion maintenance
Aluminium alloys	Medium	Moderate stiffness, good strength	High	Higher raw material cost	Larger sections needed for stiffness
Fibre-reinforced polymers (FRPs)	Low	High strength-to-weight ratio	Very high	Moderate to high	Limited long-term field data
Bio-based composites (e.g., engineered bamboo)	Low	Moderate	Variable	Low in resource-rich regions	Durability, standardization

The system-level consequences of material density are illustrated in Figure 3. Higher density increases structural mass and can consequently increase foundation requirements and tracking-system loads. Conversely, a lighter material with a higher unit price may reduce costs elsewhere in the concentrator system through lower structural mass, reduced foundation demand, easier transportation and installation, and lower tracking loads. The economic value of lightweight structural materials therefore depends on the balance between their additional material cost and the system-level savings achieved over installation and operation.

4.3 Receiver and Absorber Materials

Receiver and absorber materials govern the conversion of concentrated solar radiation into useful thermal or electrical energy and therefore have a direct influence on system efficiency and operating stability. In CSP systems, receiver materials must withstand high incident flux, steep temperature gradients, oxidation, and repeated thermal cycling, whereas CPV systems require effective heat removal to limit cell-temperature losses and maintain conversion performance [30, 31]. Material requirements consequently become more demanding as concentration ratio and operating temperature increase.

Selective coatings such as black chrome and cermet are designed to combine high solar absorptance with low thermal emittance, thereby limiting radiative heat loss from the receiver. When properly fabricated and applied, these coatings can improve thermal performance, although their long-term stability can be affected by oxidation, interdiffusion, and delamination, particularly at elevated temperatures [32, 33]. Their suitability therefore depends not only on initial optical properties but also on the retention of those properties throughout repeated thermal exposure.

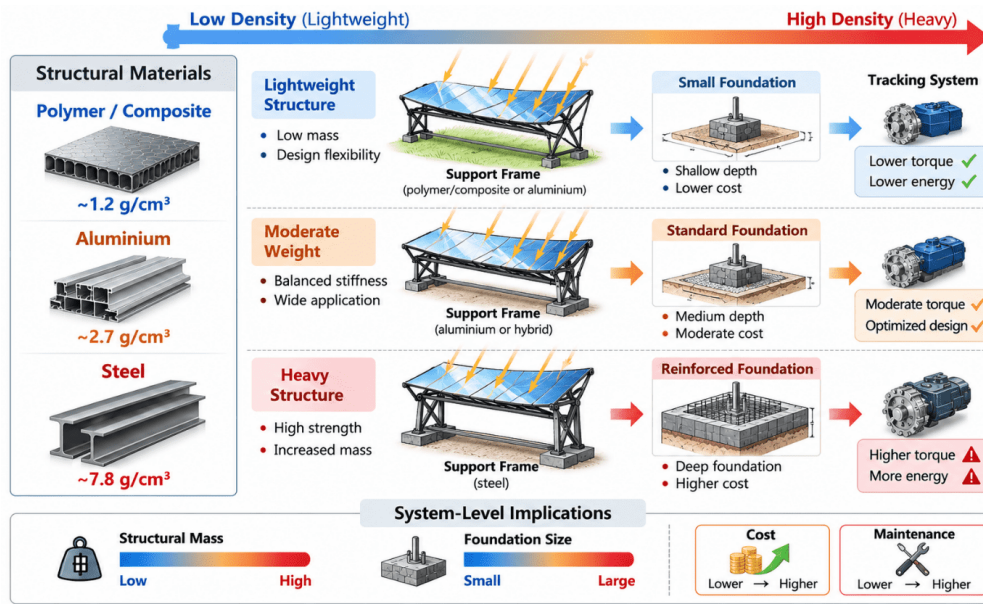


Figure 3. Schematic representation of the influence of structural material density on concentrator system design, showing its effects on structural mass, foundation size, and tracking-system load

Lower-cost alternatives include carbon-based coatings, nanostructured paints, and blackened metallic surfaces. These materials can be practical at moderate operating temperatures because of their relatively low cost and reparability, but their higher thermal emittance results in increasing radiative losses as receiver temperature rises [34, 35]. Very low-cost painted or blackened surfaces are therefore most appropriate when operating temperatures and service-life requirements remain moderate [36]. Their initial cost advantage becomes less important when thermal losses or premature degradation reduce useful energy output over the operating life of the system.

Substrate selection is equally important because the underlying material must provide mechanical integrity, thermal stability, and compatibility with the absorber coating. Stainless steels and nickel alloys can provide suitable mechanical support and coating compatibility, although they involve higher material costs and remain susceptible to oxidation at elevated temperatures. Ceramic and ceramic-matrix receivers offer greater stability at very high temperatures but introduce other constraints, particularly brittleness and fabrication complexity [33, 37]. The choice of substrate and surface treatment should therefore be considered jointly rather than as independent material decisions.

Table 4 summarizes the principal receiver and absorber options considered in this review. The comparison indicates that lower-cost absorber surfaces are more applicable to moderate-temperature systems, whereas high-temperature CSP places greater emphasis on selective surfaces and receiver materials capable of retaining their optical and thermal properties under prolonged thermal exposure. Receiver selection therefore involves a trade-off among absorptance, thermal emittance, operating-temperature capability, durability, fabrication requirements, and cost rather than a simple comparison of initial material prices.

Table 4. Comparison of receiver and absorber materials for solar concentrator applications

Receiver/Absorber Type	Typical Solar Absorptance	Thermal Emittance	Operating Temperature Range	Cost Characteristics	Key Limitations
Black-chrome coating	>0.9	Low	Up to ~400°C	Moderate to high	Limited high-temperature stability
Cermet selective coating	>0.9	Very low	Up to ~600°C	High	Complex fabrication, ageing
Carbon-based coatings	0.85–0.95	High	<400°C	Low	Higher radiative losses
Painted/blackened metals	0.8–0.9	High	<300°C	Very low	Rapid degradation
Ceramic receivers	High (with coating)	Low	>800°C	High	Brittleness, cost

4.4 Protective Coatings and Environmental Durability

Environmental exposure can offset the initial cost advantage of a material when reflectivity, mechanical integrity, or absorbance deteriorates faster than anticipated. Solar concentrator components are continuously exposed to ultraviolet radiation, thermal cycling, moisture, salts, dust, abrasion, and repeated cleaning. These stressors can accelerate optical and structural degradation and increase operating costs through soiling, corrosion, surface roughening, more frequent maintenance, and premature component replacement [38]. Environmental durability is therefore an integral part of material selection rather than a secondary consideration after initial optical or mechanical performance has been established.

Soiling is particularly important at sites with high direct normal irradiance (DNI), many of which are also exposed to substantial dust deposition. Accumulated particles reduce specular reflection, while repeated cleaning can introduce additional abrasive damage to exposed optical surfaces. Field evidence from dusty environments has shown that particulate accumulation can materially reduce solar conversion efficiency, making cleaning frequency and cleaning method relevant design and operational variables [39]. The durability of reflective surfaces must consequently be considered together with local soiling conditions and the cleaning practices required to maintain optical performance.

Corrosion protection is similarly important for silvered-glass backing layers, aluminium reflectors, steel support structures, and metallic receivers. Galvanizing, anodizing, protective paint systems, edge sealants, and multilayer barriers can extend component service life, although the adhesion, weathering resistance, and UV stability of these protective systems must themselves be qualified for the intended operating environment [40]. A protective treatment is therefore effective only when its performance remains compatible with the environmental stresses and expected service life of the underlying component.

Multifunctional coatings seek to combine several protective functions, including anti-soiling behaviour, scratch resistance, UV protection, and resistance to environmental ingress. Such coatings can reduce maintenance requirements and slow performance deterioration, although they also introduce challenges associated with fabrication, process control, and reproducibility [41, 42]. Hydrophobic anti-soiling surfaces are particularly relevant where cleaning demand is high, but their practical value depends on retaining surface functionality during prolonged outdoor weathering and repeated abrasive cleaning [43].

Table 5 compares the principal functions and limitations of the protective coatings considered in this review. The comparison shows that individual coating technologies generally address specific degradation mechanisms rather than all environmental stressors simultaneously. Anti-soiling coatings primarily target dust adhesion, corrosion-resistant systems protect metallic components, UV-stabilised coatings limit photodegradation of polymeric surfaces, and hard coatings improve resistance to scratching and abrasion. Layered or multifunctional protection can therefore be preferable when components are exposed to several interacting degradation mechanisms, provided that additional fabrication requirements and possible optical penalties are taken into account.

Table 5. Protective coating types for solar concentrator components and their primary functions

Type of Coating	Primary Function	Typical Applications	Key Benefits	Main Limitations
Anti-soiling (hydrophobic/hydrophilic)	Reduce dust adhesion	Mirrors, lenses	Lower cleaning frequency	Durability under abrasion
Corrosion-resistant coatings	Prevent oxidation/corrosion	Metal reflectors, structures	Extended service life	Added fabrication cost
Ultraviolet-stabilised coatings	Mitigate photodegradation	Polymer reflectors, lenses	Improved optical retention	Limited long-term data
Hard protective coatings (e.g., silica-based)	Improve scratch resistance	Reflective surfaces	Enhanced durability	Potential optical losses

Figure 4 illustrates the interaction between environmental stressors and the corresponding protective strategies. Dust and abrasion primarily threaten optical surfaces, while moisture and salts promote corrosion and ultraviolet exposure accelerates the degradation of susceptible polymeric materials. Protective coatings provide a barrier between these environmental stressors and the underlying component, thereby slowing the loss of optical or structural performance. Their contribution should therefore be assessed in terms of retained performance, maintenance requirements, and service-life extension rather than coating cost alone.

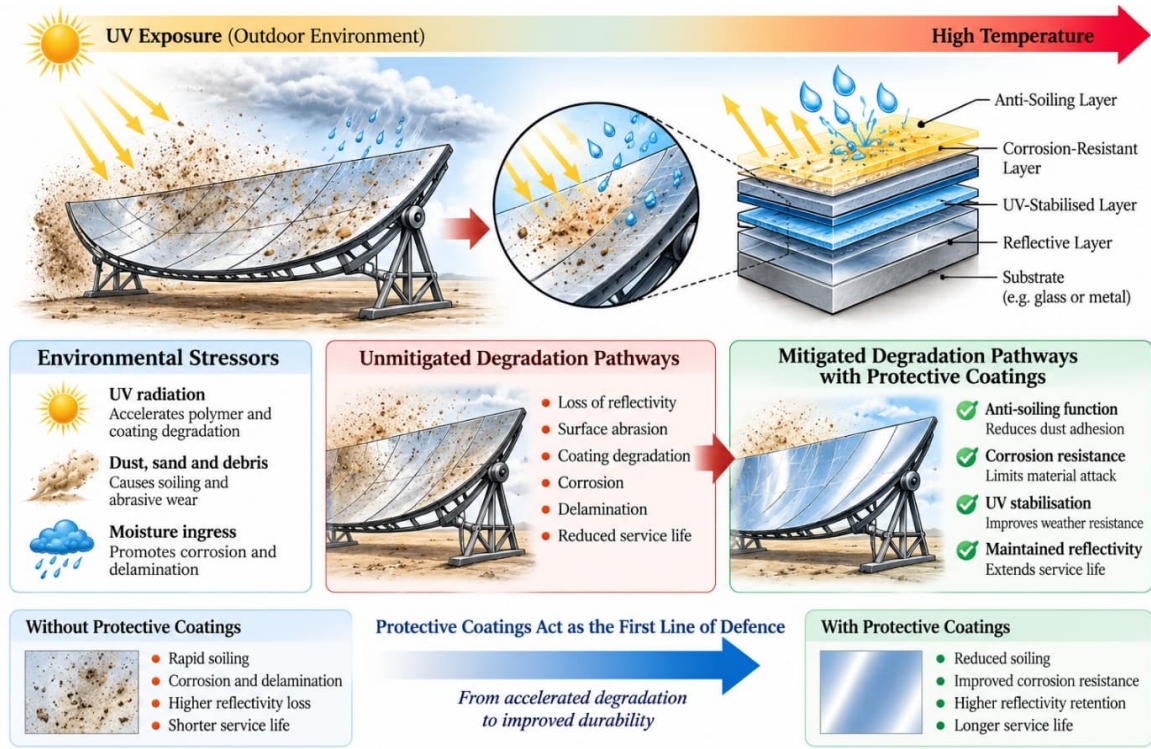


Figure 4. Schematic illustration of environmental degradation pathways in solar concentrator systems and the mitigating role of protective coatings, including anti-soiling, corrosion-resistant, and ultraviolet-stabilized layers

Note: UV = ultraviolet.

4.5 Energy Modelling, Economics, and Life-Cycle Cost

Material selection influences both energy yield and life-cycle cost. A reflector with lower initial reflectivity reduces the optical energy delivered to the receiver from the beginning of operation, while progressive degradation further decreases energy output over time. Similar effects arise from receiver deterioration, coating failure, structural misalignment, and material-related availability losses. Energy-system modelling therefore provides a useful means of translating material properties and degradation into annual energy generation, operation and maintenance (O&M) requirements, replacement needs, and LCOE. Published CSP studies using the SAM have demonstrated the value of combining resource data, optical and thermal performance, and economic assumptions within a common simulation framework [8]. Studies of hybrid energy systems have likewise shown that reliability constraints and component sizing can materially alter techno-economic conclusions [4].

4.5.1 Review-based energy-yield model

For a general review of solar-concentrator materials, a transparent reduced-order model is more appropriate than a site-specific simulation requiring assumptions regarding location, hourly weather conditions, plant layout, dispatch strategy, storage configuration, and financing. Annual net energy in year t can therefore be represented by a baseline first-year yield, E_0 , multiplied by an annual material-performance retention term. For illustrative purposes:

$$E_t = E_0(1 - d)^t \quad (1)$$

where, d denotes an annualized performance-degradation factor representing the combined influence of reflectivity loss, receiver degradation, availability reduction, and other material-related losses. Eq. (1) is not intended to replace hourly or project-specific energy simulation. Instead, it provides a simplified sensitivity framework for illustrating how persistent material deterioration can accumulate over the operating life of a solar concentrator system.

4.5.2 Conventional levelized cost of energy and material-dependent terms

The conventional discounted LCOE can be expressed as:

$$\text{LCOE} = \frac{\frac{\sum_{t=0}^N (I_t + O_t + R_t)}{(1+r)^t}}{\frac{\sum_{t=0}^N E_t}{(1+r)^t}} \quad (2)$$

where, I represents investment cost; O and M denote operating and maintenance expenditures; R represents replacement cost; r is the discount rate; and E is net electricity or useful-energy output. Material selection affects both the cost and energy terms of this expression. Reflectors and structural materials influence initial investment, protective coatings influence cleaning and replacement requirements, and material degradation affects lifetime energy production.

Life-cycle evidence indicates that long-term optical retention and structural durability can be more important than modest reductions in initial material cost [44, 45]. This is particularly relevant when lower-cost materials require more frequent replacement, greater cleaning effort, or additional downtime. A material that appears attractive at the point of purchase may therefore become less economical when degradation-related energy losses and maintenance requirements are considered over the full project life.

Table 6 summarizes the principal economic pathways through which different material categories influence system cost and performance. Reflective materials primarily affect optical retention and cleaning or replacement requirements, structural materials influence installed mass and tracking demand, receiver and absorber materials affect thermal efficiency, and protective coatings can influence maintenance intensity and service-life extension. Protective systems may represent only a modest share of initial capital expenditure while exerting a larger influence on O&M requirements and lifetime performance.

Table 6. Economic impacts of material choices in large-scale solar concentrator systems

Material Category	Impact of Capital Cost	Impact of Operational Cost	Key Economic Trade-off
Reflective materials	High	Medium to high (cleaning, replacement)	Upfront cost vs. reflectivity retention
Structural materials	Medium to high	Medium (maintenance, tracking energy)	Weight reduction vs. stiffness and durability
Receiver/Absorber materials	Medium	Low to medium (efficiency losses)	Efficiency gains vs. coating cost
Protective coatings	Low to medium	High (maintenance reduction)	Added cost vs. lifetime extension

Figure 5 illustrates the resulting life-cycle trade-off between initial material cost, durability, cumulative maintenance and replacement expenditure, and retained energy output. A moderate increase in initial cost can be economically justified when it reduces subsequent maintenance or replacement requirements and preserves energy production over the operating life of the system.

4.5.3 Illustrative sensitivity analysis

A one-at-a-time sensitivity framework was applied to the normalized life-cycle model to examine the direction and relative importance of material-related decision variables. The baseline was deliberately dimensionless: first-year energy output was set to 1.0, initial material-related capital cost to 1.0, annual material-related O&M cost to 0.02, mid-life replacement allowance to 0.20, project life to 25 years, annual degradation to 0.5%, and discount rate to 7%. The analysis was intended to illustrate relative sensitivity rather than to predict the performance or economics of a specific plant.

Table 7 presents the normalized results. The analysis shows that changes in material-related capital cost, degradation rate, O&M expenditure, discount rate, project life, and replacement assumptions can all shift the normalized LCOE index. In particular, higher degradation, greater replacement burden, and less favourable financing assumptions increased the life-cycle cost index, whereas lower O&M requirements and longer operating life reduced it. These results support evaluating material options on a life-cycle basis rather than relying on purchase price alone.

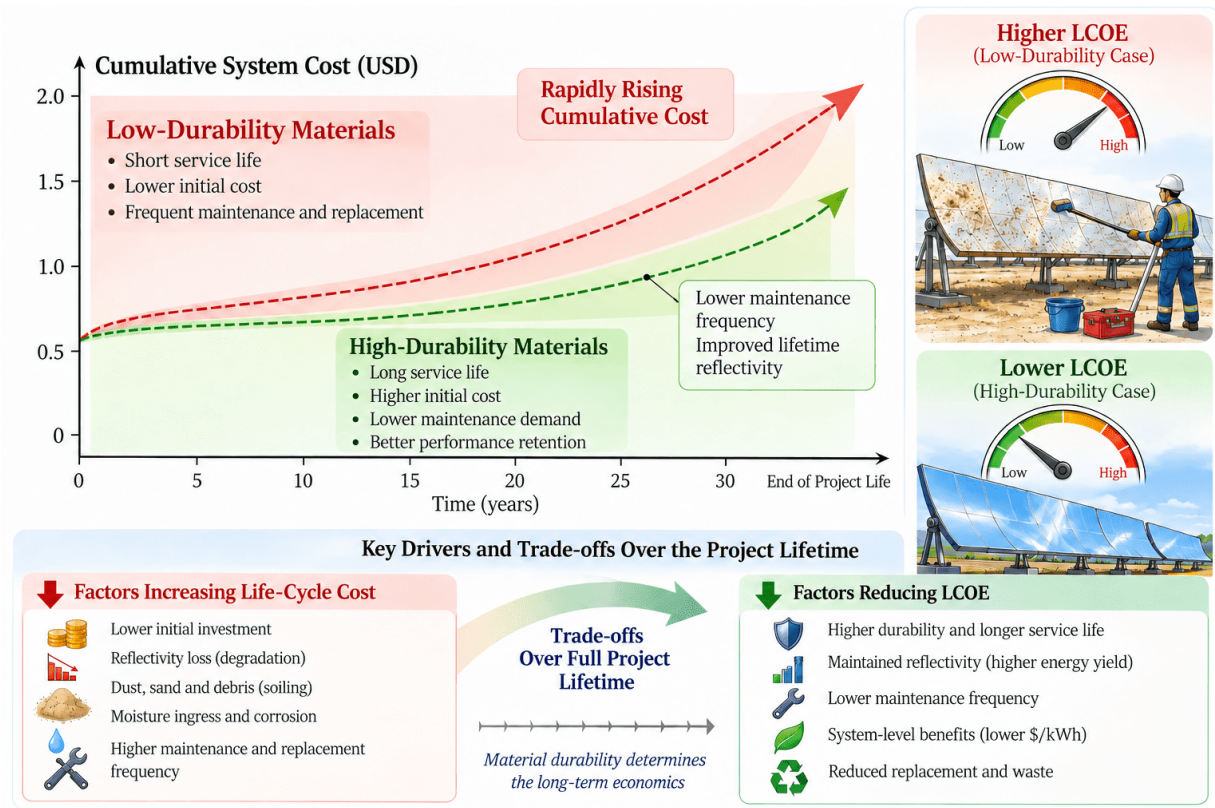


Figure 5. Conceptual relationship among material durability, cumulative system cost, and levelized cost of energy (LCOE) for solar concentrator systems

Table 7. Illustrative normalized sensitivity of material-related levelized cost of energy (LCOE) to selected assumptions

Parameter	Case	Normalized LCOE Index (baseline = 1.00)	Interpretation
Initial material CAPEX	−20%	0.85	Lower lifetime cost
Initial material CAPEX	+20%	1.15	Higher lifetime cost
Annual degradation	0.25%	0.98	Limited change
Annual degradation	2.0%	1.13	Higher lifetime cost
Annual material O&M	−50%	0.91	Lower lifetime cost
Annual material O&M	+50%	1.09	Higher lifetime cost
Discount rate	4%	0.82	Lower lifetime cost
Discount rate	10%	1.20	Higher lifetime cost
Project life	20 years	1.08	Higher lifetime cost
Project life	30 years	0.95	Lower lifetime cost
Mid-life replacement	None	0.93	Lower lifetime cost
Mid-life replacement	40% of initial	1.07	Higher lifetime cost

Note: CAPEX = capital expenditure; O&M = operation and maintenance.

Sensitivity analysis is also relevant when material decisions interact with reliability or environmental constraints. In hybrid renewable-energy systems, loss-of-power-supply probability has been shown to influence both system sizing and cost, demonstrating that reliability requirements can alter apparently least-cost solutions [44]. A similar principle applies to solar concentrator materials. A reflector with a low initial cost but a high degradation rate may become unattractive when minimum energy-output, availability, maintenance, or replacement requirements are imposed. Material selection should therefore account for both economic sensitivity and the operational constraints that determine whether nominal cost savings can be sustained over the project life.

4.6 Environmental, Carbon, and Societal Implications

4.6.1 Life-cycle carbon emissions and environmental cost

The transition to solar energy reduces operational dependence on fossil-fuel generation, but the production, transport, maintenance, and replacement of concentrator materials still involve embodied emissions. The environmental performance of a material should therefore be considered over its life cycle rather than solely during system operation. Studies of grid-related emissions have shown that the reference electricity mix must be clearly defined when estimating avoided CO₂ emissions because the emissions avoided per unit of solar energy depend on the generation source being displaced [4]. Similarly, assessments across different fuel types have demonstrated that energy, economic, and environmental outcomes change with the choice of reference technology [5]. For life-cycle decision making, an environmental damage term can be incorporated into the conventional LCOE formulation. A generalized environmentally adjusted metric can be expressed as:

$$\text{LCOE}_{\text{env}} = \frac{\sum_{t=0}^N \frac{(I_t + O_t + M_t + R_t + C_{\text{CO}_2,t})}{(1+r)^t}}{\sum_{t=0}^N \frac{(E_t)}{(1+r)^t}} \quad (3)$$

$$C_{\text{CO}_2,t} = \frac{m_{\text{CO}_2,t}}{100} \times SCC \quad (4)$$

where, $m_{\text{CO}_2,t}$ denotes the attributable CO₂ emissions in year (t), expressed in kilograms, and SCC represents the assumed social cost of carbon, expressed in currency per tonne of CO₂. Incorporating carbon-damage valuation into renewable-energy economic assessment makes environmental externalities explicit within the cost framework rather than treating them separately from the economic comparison [8, 9]. Life-cycle carbon-accounting studies have also emphasized the need for consistent system boundaries and emission factors to ensure meaningful comparison across technologies [9].

The purpose of this environmentally adjusted metric is not to assign a universal monetary value to the environmental performance of a particular concentrator material. Rather, it provides a transparent framework for examining how embodied emissions, material replacement, retained energy yield, and the assumed carbon value can influence life-cycle comparisons. Materials with higher embodied impacts may remain attractive when longer service life or better performance retention reduces replacement demand and preserves lifetime energy output, whereas an initially low-impact material may lose part of that advantage if frequent replacement is required.

4.6.2 Social and wider environmental effects

Large-scale solar deployment has environmental and societal implications that extend beyond carbon emissions. Relevant considerations include land occupation, water consumption, visual impact, local employment, worker safety, supply-chain localization, recyclability, waste generation, and community acceptance. Photovoltaic case studies have shown that low-carbon electricity generation still requires explicit consideration of environmental and societal effects at the project level [5]. Approaches that assign monetary values to social and environmental externalities have also been applied to wind-energy projects, providing a transferable framework for broader renewable-energy assessment [15].

For solar concentrator systems, many of these effects are directly related to material choice. Heavy steel and glass components can increase transport requirements and embodied-energy burdens, whereas longer-lasting components can reduce material replacement and associated waste. Locally manufacturable structures may contribute to regional economic activity and shorter supply chains, but these advantages depend on maintaining the required manufacturing quality. Water-intensive cleaning can create additional constraints in arid regions, while end-of-life management of polymers and composite materials may be more difficult than the recycling of established metallic materials. Broader life-cycle studies of solar-powered systems reinforce the importance of accounting for upstream production and end-of-life burdens rather than focusing exclusively on operational energy use [46].

Table 8 provides a practical set of indicators for incorporating environmental and social considerations into material-selection decisions. The proposed indicators connect material characteristics with embodied carbon, operational CO₂ avoidance, water demand, land and visual impacts, circularity, and local socio-economic value. Where possible, material burdens should be related to a functional measure of system performance, such as aperture area or lifetime energy output, rather than to component mass alone. This approach allows durability and retained energy performance to be considered alongside the environmental burden associated with producing, maintaining, and replacing a material.

Studies of hybrid renewable-energy systems have shown that environmental and economic objectives should be considered together, particularly when storage, reliability, and actual operating conditions influence system performance [5]. The same principle supports a multi-criteria approach to solar-concentrator material selection in

which initial cost is considered alongside durability, lifetime energy yield, maintenance demand, carbon implications, resource use, and end-of-life performance. Such an approach is more consistent with life-cycle decision making than selecting materials solely on the basis of minimum purchase price.

Table 8. Environmental and social indicators relevant to material selection for solar concentrator systems

Dimension	Material Linkage	Preferred Assessment Indicator	Implications on Decision
Embodied carbon	Glass, steel, aluminium, composites, coatings	kg CO ₂ -eq per m ² aperture or per kWh lifetime output	Favours durable and low-carbon supply chains
Operational CO ₂ avoidance	Lifetime energy yield and displaced grid mix	<i>t</i> CO ₂ avoided over project life	Higher retained yield increases benefit
Water	Cleaning frequency and coating performance	<i>L</i> water per MWh or per m ² -year	Anti-soiling surfaces valuable in arid regions
Land/Visual impact	Concentrator geometry and structural footprint	ha/MW and qualitative visual assessment	Site-specific planning required
Circularity	Repairability, recyclability, replacement interval	Recycled content; recovery rate; replacement count	Favours long-lived recyclable components
Local socio-economic value	Local materials, fabrication, maintenance	Local content; jobs; supply-chain share	Supports regional manufacturing when quality could be assured

4.7 Emerging Trends, Research Gaps, and Deployment Roadmap

Material development for solar concentrator systems is increasingly directed toward thinner glass mirrors, hybrid metal–polymer reflectors, multifunctional coatings, modular lightweight structures, and regionally sourced materials. These developments seek to reduce transportation and structural burdens, foundation requirements, maintenance demand, and replacement costs while preserving optical performance over extended operating periods [11]. Their practical value, however, depends on whether improvements demonstrated at the material or component level can be retained under large-scale manufacturing and long-term outdoor exposure.

Multifunctional surfaces that combine anti-soiling properties, corrosion resistance, UV protection, and abrasion resistance are particularly relevant because maintenance requirements increase with collector area. Combining several protective functions within a single surface system may reduce cleaning and replacement demand, but laboratory-scale performance must ultimately be translated into reproducible manufacturing processes supported by quality control and field validation [41]. Long-term performance retention is therefore as important as initial coating functionality when evaluating these technologies for large-scale deployment.

Table 9 summarizes the principal emerging material trends and the research gaps that currently limit their wider application. Across these technologies, the main challenge is not necessarily achieving favourable initial laboratory performance. Greater uncertainty lies in long-term durability, manufacturing consistency, repairability, scalability, and the extent to which accelerated testing can predict performance under actual operating conditions.

Table 9. Emerging material trends and associated research gaps in solar concentrator systems

Emerging Trend	Potential Benefits	Current Limitations	Key Research Gaps
Ultra-thin glass mirrors	Reduced weight, high reflectivity	Handling fragility	Long-term field durability
Hybrid polymer–metal reflectors	Low cost, lightweight	Delamination risk	Moisture and ultraviolet resistance
Multifunctional coatings	Reduced maintenance, improved durability	Complex fabrication	Scalable manufacturing methods
Bio-based structural materials	Low cost, sustainability	Limited standards	Long-term mechanical behaviour

The gap between accelerated ageing and long-term field exposure is illustrated in Figure 6. Laboratory tests

can isolate individual degradation mechanisms and shorten evaluation periods, but they may not fully reproduce the combined effects of ultraviolet radiation, dust, humidity, salts, thermal cycling, and abrasion caused by repeated cleaning. Lifetime predictions for emerging low-cost materials should therefore be interpreted cautiously until accelerated-ageing results are supported by sufficiently representative field evidence.

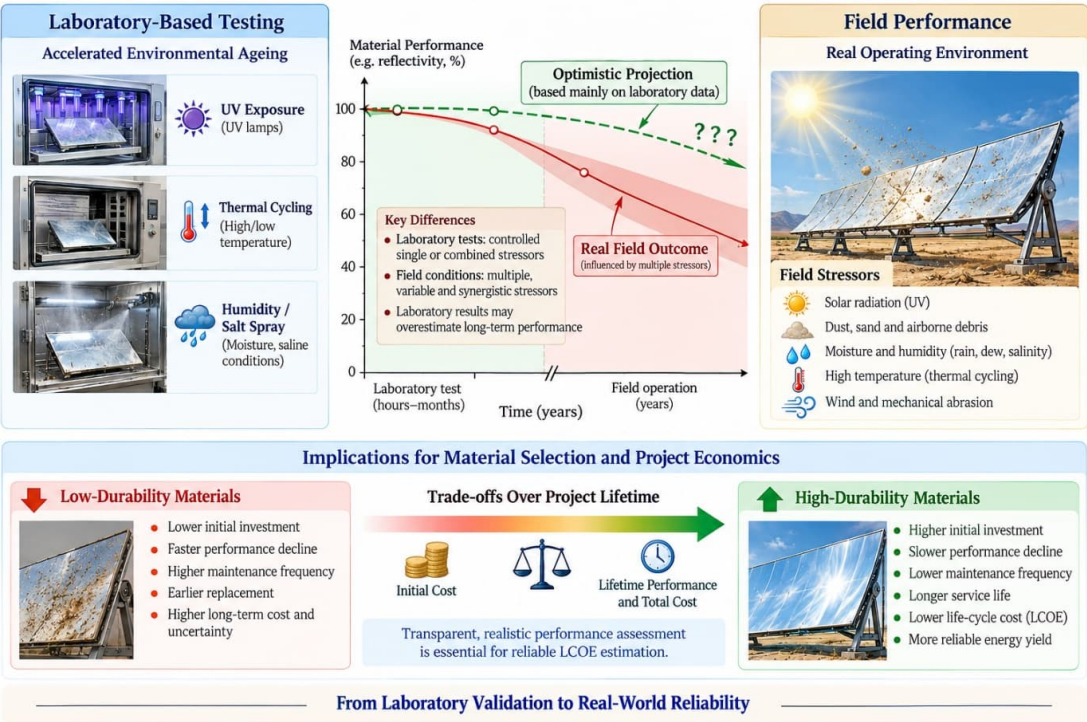


Figure 6. Conceptual comparison of laboratory-based accelerated ageing tests and long-term field performance of solar concentrator materials, illustrating uncertainty in lifetime prediction for emerging low-cost materials

Note: UV = ultraviolet; LCOE = levelized cost of energy.

4.8 Material Qualification and Deployment Timeline

A practical pathway from material development to large-scale solar-concentrator deployment should distinguish material qualification from the broader transition of the energy system. For individual materials and components, qualification can proceed through successive stages of laboratory screening, accelerated ageing, multi-climate field exposure, pilot manufacturing, and commercial demonstration. These stages address different sources of uncertainty, progressing from basic material performance to durability, manufacturing reproducibility, field reliability, and ultimately commercial bankability. National or global solar-transition budgets cannot be derived credibly from the evidence reviewed here because they depend on target capacity, grid expansion, energy storage, financing conditions, labour, land availability, and country-specific policy. The material-development process can nevertheless be organized into indicative qualification stages and associated cost categories.

Table 10 presents an indicative material-qualification pathway from initial screening to bankable deployment. The proposed time ranges are intended to show the approximate sequence and possible overlap of qualification activities rather than fixed development schedules. Early stages concentrate on laboratory testing and accelerated ageing, whereas later stages require progressively greater investment in outdoor validation, pilot manufacturing, process control, commercial demonstration, warranties, and long-term monitoring. Advancement between stages should depend on whether the material satisfies the corresponding technical and durability requirements rather than on elapsed time alone.

Experience from national green-economy transition studies can provide useful examples of how broader implementation schedules and investment requirements may be structured once a specific geography, demand trajectory, and policy objective have been defined. Such planning should, however, be undertaken separately for individual countries and projects. Generic material-performance evidence alone is insufficient to determine national deployment schedules or investment requirements because resource conditions, infrastructure, financing, supply chains, labour availability, and policy settings vary substantially across locations.

Table 10. Indicative material-qualification timeline and cost categories for the transition from research to bankable solar-concentrator deployment

Phase	Indicative Timing	Primary Activities	Budget Requirement/Cost Category	Decision Gate
1. Screening	0–2 years	Optical, thermal, mechanical and corrosion screening; coupon tests	Research and development staff, laboratory equipment, prototype materials	Meets minimum optical-mechanical thresholds
2. Accelerated qualification	1–3 years	Ultraviolet, humidity, salt, abrasion, thermal cycling; failure analysis	Environmental chambers, metrology, coating/process development	Degradation models are repeatable
3. Multi-climate field validation	2–5 years	Outdoor exposure in dusty, tropical, coastal, and temperate sites	Test racks, instrumentation, cleaning trials, logistics	Field retention matches qualification model
4. Pilot manufacturing	3–6 years	Scale-up, Quality assurance/quality control, joining, installation and repair procedures	Pilot tooling, process control, supply-chain qualification	Reproducible product at target cost
5. Bankable deployment	5–10 years	Commercial demonstration and fleet monitoring	Demonstration capital expenditure, warranties, spares, monitoring	Lifetime risk acceptable to developers/financiers

5 Conclusions

This review shows that cost-effective material selection for large-scale solar concentrators cannot be based on purchase price alone. Silvered glass, galvanized steel, and advanced selective coatings remain important reference technologies because of their established performance and comparatively well-documented durability. Aluminium reflectors, polymer films, fibre-reinforced composites, bio-based structures, and lower-cost absorber materials can reduce structural mass or initial expenditure, but their suitability depends on whether these advantages are retained under the required operating temperature, environmental exposure, and service-life conditions. Material selection is therefore inherently application specific and should be based on lifetime performance rather than initial cost in isolation.

The life-cycle framework developed in this review clarifies the connection between material properties, energy performance, and economic outcomes. Material choice affects the cost side of LCOE through investment, maintenance, cleaning, and replacement, while degradation of reflectors, receivers, coatings, and structural components can reduce lifetime energy output. The illustrative sensitivity analysis further shows that degradation rate, O&M requirements, replacement assumptions, discount rate, and project lifetime can materially influence life-cycle comparisons. These variables should therefore be reported explicitly when lower-cost materials are evaluated against more durable alternatives. In particular, reductions in initial expenditure should not be interpreted as life-cycle savings unless the associated effects on performance retention, maintenance, replacement, and energy yield are also considered.

Environmental conditions are equally important to material qualification. Dust, humidity, salinity, ultraviolet radiation, thermal cycling, and cleaning-related abrasion produce different degradation pathways and can alter the relative suitability of competing materials. Material-performance claims should therefore be linked to clearly defined exposure conditions and, where possible, supported by multi-climate field validation. Site-specific energy assessment should likewise rely on appropriate resource data and account for uncertainty in resource modelling when estimating energy yield and LCOE. Environmental evaluation should extend beyond operational emissions to include embodied carbon, avoided grid emissions, cleaning-water requirements, land implications, recyclability, replacement waste, local supply chains, and carbon externalities. In this context, durability has both economic and environmental significance because longer service life and retained performance can reduce replacement demand while preserving

lifetime energy production.

The evidence reviewed also identifies a persistent gap between promising laboratory performance and commercially reliable field operation. Emerging lightweight reflectors, multifunctional coatings, composite structures, and other lower-cost materials require stronger evidence of long-term durability, manufacturing consistency, reparability, and performance under interacting environmental stressors. Accelerated ageing is useful for screening and qualification, but it should be complemented by representative field exposure before long-term performance or economic claims are generalized. A staged pathway from laboratory screening and accelerated testing to multi-climate validation, pilot manufacturing, and commercial demonstration provides a practical basis for reducing this uncertainty.

Large-scale deployment should therefore be approached as a system-level material-selection problem rather than as a search for a single universally low-cost material. High-retention optics, lightweight but sufficiently stiff structures, temperature-appropriate receivers, durable protective surfaces, standardized qualification procedures, climate-specific field validation, reproducible manufacturing, and transparent techno-economic and environmental assessment must be considered together. The central implication of this review is that future reductions in the cost and environmental burden of CSP and CPV systems are more likely to result from coordinated improvements across these interacting subsystems than from minimizing the cost of any individual material.

Author Contributions

Conceptualization, A.U. and D.K.; methodology, A.U. and D.K.; software, A.U. and D.K.; validation, A.U. and D.K.; formal analysis, A.U. and D.K.; investigation, A.U. and D.K.; resources, A.U.; writing—original draft preparation, A.U. and D.K.; writing—review and editing, A.U. and D.K.; visualization, A.U. and D.K.; All authors have read and agreed to the published version of the manuscript.

Data Availability

Not applicable. This manuscript does not report data generation or analysis.

Conflicts of Interest

The authors declare no conflicts of interest.

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